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Editorial Office
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Dear Editors: Profile-Driven Hardware Simulation for Organic Optoelectronic Edge Vision

Article Type: Article

Editorial Summary

We present an open-source simulation framework for organic optoelectronic compute-in-memory inference with modern vision transformers. Our central contribution, Ensemble HAT, resamples device-to-device mismatch masks every epoch. On Tiny-ViT-5M, it raises fresh-instance accuracy from chance level (10.00%) to 86.37% (mean $\pm$ 1.54%), enabling robust deployment without per-device calibration.

We rigorously falsify three intuitive strategies for breaking the severe nonlinearity ceiling—joint linearization training, all-linear ablations, and MLP-only controls—all converging to an approximately 30% fresh-instance limit. Rather than weakening the study, this negative result exposes a structural generalization barrier in the attention pathway that first-order behavioral surrogates cannot circumvent.

Novelty Claims

- We introduce the first open-source framework for organic optoelectronic CIM with vision transformers, providing PyTorch-native device-profile substitution and hybrid analog/digital deployment.
- We identify hardware-instance overfitting in organic CIM inference and propose Ensemble HAT, a training-time mismatch-mask resampling fix that restores generalization.
- We falsify three severe-nonlinearity mitigation strategies, establishing an approximately 30% structural ceiling that defines the boundary of first-order behavioral simulation.

Audience

Nature Communications readers will find a reusable methodology for evaluating organic device assumptions on vision tasks and a rigorous negative-result anchor that delineates the limits of first-order behavioral simulation.

Related Submissions

This work is not under consideration elsewhere.

Suggested Reviewers

[To be provided]

Contact Information

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Sincerely,

[Corresponding author name]